

Salmonella Enteritidis PT59 Outbreak

Overview

This dataset describes a food-borne outbreak of *Salmonella* Enteritidis phage type 59 (PT59). The data include a food distribution network represented as a directed graph of supplier-retailer links, and genetic cluster assignments for isolates, enabling joint epidemiological and molecular analysis of the outbreak.

Source

Multi-agency Outbreak Control Team (Public Health England).

Data

The dataset is returned as a `NamedTuple` with two `DataFrames`:

- `graph`: from, to (edges of the food distribution network)
- `cluster`: genetic cluster assignments for isolates

```
using Outbreaks
using DataFrames
data = Outbreaks.s_enteritidis_pt59()
graph = data.graph
cluster = data.cluster
```

	Column1	obj\$cluster
	String7	String1
1	d81c17	A
2	064974	A
3	3b712b	A
4	486c07	A
5	6f5824	A
6	c77a84	A
7	1f4d22	A
8	f951d8	A
9	44060b	A
10	905296	A
11	b4e5d5	A
12	0fffca	B
13	78e5ba	B
14	e45c54	B
15	ca432a	B
16	0b6e5a	B
17	ef7028	B
18	732379	B
19	82b4fc	B
20	9199be	B
21	cede47	B
22	9f5aad	C
23	acf7bb	C
24	24876f	C
...	...	...

Distribution network

```
first(graph, 6)
```

	Column1	from	to
	Int64	String7	String7
1	1	2e7967	dd73b6
2	2	2e7967	c7cd02
3	3	2e7967	2afba4
4	4	2e7967	4df851
5	5	2e7967	48f980
6	6	2e7967	2d3187

```
describe(graph)
```

	variable	mean	min	median	max	nmissing	eltype
	Symbol	Union...	Any	Union...	Any	Int64	DataType
1	Column1	52.0	1	52.0	103	0	Int64
2	from		030327		fc7f8f	0	String7
3	to		030327		f951d8	0	String7

Genetic clusters

```
first(cluster, 6)
```

	Column1	obj\$cluster
	String7	String1
1	d81c17	A
2	064974	A
3	3b712b	A
4	486c07	A
5	6f5824	A
6	c77a84	A

```
describe(cluster)
```

	variable	mean	min	median	max	nmissing	eltype
	Symbol	Nothing	InlineSt...	Nothing	InlineSt...	Int64	DataType
1	Column1		064974		f951d8	0	String7
2	obj\$cluster		A		C	0	String1